

Quantitative Data Analysis

Part 2 of the Research Methods Professional Certificate · 10 units · 25 contact hours · Intermediate

Jesse Anak · Institute of Computational Biology and Bioinformatics

What to do once the data are collected. This part starts where real analysis starts — a messy spreadsheet — and works through cleaning, missing data, description, inference and regression modelling to a written results section. The emphasis throughout is on choosing the right method for your design and interpreting what comes out, not on memorising formulas. Every technique is demonstrated in both SPSS and R, so you can work in whichever your institution uses.

Who it is for

- Researchers with data collected and no clear analysis plan
 - Postgraduate students at the analysis stage of a thesis
 - Monitoring and evaluation officers working with survey data
 - Clinicians and laboratory scientists analysing their own studies
 - Anyone reporting numbers who wants to stop guessing at the statistics
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Prerequisites

- Part 1, or an equivalent understanding of research design and sampling
 - Comfortable working with a spreadsheet
 - No programming experience assumed — R is taught from the first command
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Learning outcomes

- Build a codebook and turn completed questionnaires into an analysable dataset
- Screen data for impossible values, duplicates and inconsistencies
- Choose a defensible strategy for missing data and report it
- Summarise variables honestly and build a publication-ready Table 1
- Produce figures that show the data rather than hiding them
- Interpret p-values, confidence intervals and effect sizes accurately
- Select the correct test for your design, outcome and assumptions
- Fit and diagnose linear and logistic regression models
- Distinguish confounders, mediators and colliders when choosing covariates

- Write a results section that matches the analysis you actually ran

Schedule of units

TEN UNITS

#	Unit	Duration	Focus
1	From Questionnaire to Dataset	120 minutes	Turning completed forms into a rectangular dataset a computer can read: the codebook, variable naming, data entry, and the decisions that are expensive to reverse later.
2	Cleaning and Preparing Data	150 minutes	Finding what is wrong with a real dataset — impossible values, duplicates, inconsistent categories — and building the derived variables the analysis actually needs, all as a script you can rerun.
3	Missing Data	120 minutes	Why data are missing determines what you may do about it. Deletion, single imputation and multiple imputation, and what each assumes.
4	Describing Data and Building Table 1	120 minutes	Summarising variables honestly: which measure of centre and spread fits which distribution, and how to assemble the participant characteristics table that opens almost every quantitative paper.
5	Visualising Data	120 minutes	Choosing a chart that shows what the data actually do, and avoiding the plots that hide the very thing a reader needs to see.
6	The Logic of Inference	150 minutes	What a p-value is and is not, what a confidence interval actually tells you, and why an effect size belongs in every result you report.
7	Comparing Groups	180 minutes	Choosing the right test for your design, outcome and assumptions — t-tests, ANOVA, chi-square and their non-parametric alternatives — and running them in SPSS and R.
8	Correlation and Linear Regression	180 minutes	Modelling a continuous outcome: what a regression coefficient means, how to handle categorical predictors, and how to check the model is entitled to the conclusions you want from it.
9	Logistic Regression and Confounding	180 minutes	Modelling binary outcomes: odds, odds ratios and what they do and do not mean, plus adjusting for confounding and recognising mediators and interactions.
10	Reporting, Reproducibility and the Results Section	150 minutes	Turning output into a results section a reviewer will accept: what to report, how to lay out tables, and how to make the whole analysis reproducible by someone else.

Assessment

Assessment is continuous and practical, ending in an analysis of a supplied dataset written up as a short results section.

COMPONENTS

Component	Weight	Detail
Unit quizzes	30%	A quiz after each of the ten units, with immediate feedback and an explanation for every option. Unlimited attempts.
Analysis labs	30%	A practical task per unit carried out on a real dataset in SPSS or R, reviewed during the live session.
Final analysis report	40%	A complete analysis of a supplied dataset: cleaning decisions, Table 1, a regression model with diagnostics, and a written results section. Marked against the rubric below with written feedback.

FINAL PROPOSAL MARKING RUBRIC

Criterion	Weight	What earns the marks
Data preparation and cleaning	15%	Checks documented; decisions justified; raw data preserved.
Missing data handling	15%	Mechanism considered; method appropriate and reported with analysis n.
Descriptive statistics	15%	Summaries match distribution shape; Table 1 complete and correctly formatted.
Choice of test or model	25%	Method fits design and outcome; assumptions checked; alternatives considered.
Interpretation	20%	Estimates, intervals and effect sizes read correctly in the units of the problem.
Reporting and reproducibility	10%	Results section complete and free of interpretation; analysis rerunnable from script.

Pass mark: 60% overall.

Certificate

ICBB Certificate of Completion — Quantitative Data Analysis (Part 2). Awarded on achieving 60% overall and submitting the final analysis report.

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